



How to calculate $[10^{10^{10^{10^{10^10}}}} - 10^{10}]?$

Asked 14 years, 6 months ago Modified 1 year, 11 months ago Viewed 7k times



How to find an integer part of $10^{10^{10^{10^10}}}$? It looks like it is slightly above $10^{10^{10}}$.

45

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edited Oct 27, 2021 at 16:19

asked Oct 27, 2011 at 0:18



GH from MO



Vladimir Reshetnikov

- 2 Only slightly related, but for amusement, check out math.stackexchange.com/questions/72646/...
– Joel David Hamkins Oct 27, 2011 at 0:27
- 4 @quid: It is -10^{10} , not $(-10)^{10}$. – GH from MO Oct 27, 2011 at 0:28
- 1 @GH: Thanks! So, it was me being confused. Sorry for the noise. – user9072 Oct 27, 2011 at 0:35
- 3 Just for curiosisty: Was there any context for the appearance of this number? – efs Oct 27, 2021 at 16:26
- 2 @EFinat-S I experimented with various power towers, just for fun, and then I thought: is it possible to construct a non-trivial problem about power towers that results in a "small" number (i.e. a number whose decimal expansion is practically possible to fully write down), and I came up with this question.
– Vladimir Reshetnikov Oct 27, 2021 at 23:05 [Delete](#)

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1 Answer

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94

I think the number in question is $10^{10^{10}} + 10^{11} \ln^4(10)$ plus a tiny positive number. That is, it starts with a digit 1, followed by $10^{10} - 13$ zeros, then by the string 2811012357389, then a decimal point, and then some garbage (which starts like 4407116278...).



To see this let $x := 10^{-10^{10}}$, a tiny positive number, and put $c := \ln(10)$, an important constant. We have





$$10^{10^x} = 10^{1+cx+O(x^2)} = 10 + 10c^2x + O(x^2)$$

$$10^{10^{10^x}} = 10^{10+10c^2x+O(x^2)} = 10^{10} + 10^{11}c^3x + O(x^2)$$

$$10^{10^{10^{10^x}}} = 10^{10^{10} + 10^{11} c^3 x + O(x^2)} = 10^{10^{10}} + 10^{10^{10}} 10^{11} c^4 x + O(x^2),$$

where $O(x^2)$ means something tiny all the way.

In the last expression we have $10^{10^{10}} 10^{11} c^4 x = 10^{11} c^4$, which justifies my claim.

edited Oct 27, 2011 at 1:09

answered Oct 27, 2011 at 1:03



GH from MO

- 35  I'm not convinced the question belongs on MO, but I like the answer too much to vote the question down. – [Gerry Myerson](#) Oct 27, 2011 at 4:45

3  @Gerry: I thought the same! :-)) – [GH from MO](#) Oct 27, 2011 at 14:39

1  Robert Munafo has a "hypercalc", unfortunately not for the windows-environment, so I cannot use it (Actually it is a perl-script). mrob.com/pub/perl/index.html cite: "Hypercalc: An unusual calculator program. It represents numbers in a special way allowing the calculation of quantities much larger than tools such as bc, dc, MACSYMA/maxima, Mathematica and Maple, all of which use a bignum library. For example, you can use Hypercalc to determine whether $128^{48^{1024}}$ is larger than $8^{88^{888}}$. (...)" (from R. Munafo's site). Someone who could try it with this program? – [Gottfried Helms](#) Oct 27, 2011 at 17:41

4  As far as I can tell, hypercalc just uses standard floating point representations of iterated logs. This suggests that it cannot handle the sort of precision necessary here. – [S. Carnahan](#) ♦ Oct 28, 2011 at 3:55